

WATERWATCH · DESIGN DECISION RECORD

WaterWatch TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WWT-DDR-001	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/waterwatch		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D10; items O1 to O3 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed ten items as "Proposed, awaiting Amish", and the design precis WWT-PRC-001 v0.2 listed key design choices, each with options and a recommendation. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open. The same instruction approved portfolio-wide decisions on the SwapCell interface (v0.3), on pricing shared SwapCell packs once, and on community designs picking co-design partners per area later. Only the last applies to WaterWatch.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in WWT-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Chlorine sensor	Option (a): a membrane-free graphite three-electrode sensor with a potentiostat front end, about \$25, with a monthly DPD comparison built into the maintenance routine. Decided by Amish, 2026-09-25: go with recommendation.
D2	pH	Option (a): site pH entered at installation and checked monthly, no added cost. A pH probe (option b) is kept as a documented variant for sites whose pH varies. <code>budget_usd</code> stays at \$300. Decided by Amish, 2026-09-25: go with recommendation.
D3	Connectivity	Cellular LTE-M and NB-IoT with 2G fallback and SMS alerts for the first build, with the modem as a swappable module so a LoRaWAN module is a variant. This keeps the "GSM or LoRa" pitch. Decided by Amish, 2026-09-25: go with recommendation.
D4	Mounting	A separate pole beside the tapstand, not on it. Decided by Amish, 2026-09-25: go with recommendation.
D5	Sampling	An hourly 90 s timed flush through a latching valve, not continuous flow. Decided by Amish, 2026-09-25: go with recommendation.
D6	Default alert thresholds	Free chlorine below 0.2 mg/L and turbidity above 5 NTU, confirmed by two consecutive readings, adjustable per site. Decided by Amish, 2026-09-25: go with recommendation.
D7	Drain destination	The tapstand basin by default; a soakaway or a container for non-drinking use where users prefer, decided per site. Decided by Amish, 2026-09-25: go with recommendation.
D8	First site type	Tapstands first; tank outlets and kiosks later. Decided by Amish, 2026-09-25: go with recommendation.
D9	Battery	LiFePO4, with charging blocked below 0 °C and above 45 °C, rather than lithium-ion 18650 cells. Decided by Amish, 2026-09-25: go with recommendation.
D10	SwapCell	Not used. WaterWatch needs about 0.5 Wh per day and a 48 V SwapCell pack does not fit. The SwapCell interface v0.3 items therefore do not apply. Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items that remain open.

#	Item	Status
O1	First co-design partner and region. The TRL 2 note suggested approaching a maintenance service provider, but Amish directed that community designs pick co-design partners per area later, so no partner, partner type or region is chosen here.	Proposed, awaiting Amish
O2	Alert language, recipients and wording beyond the R13 rule; to come from caretakers and operators in co-design. No recommendation was made.	Proposed, awaiting Amish
O3	Who owns the data and who may see it (co-design checklist). No recommendation was made.	Proposed, awaiting Amish

■ Consequences

- `project.yaml` : unchanged apart from the TRL fields. No budget, pitch or problem change was recommended, so `budget_usd` stays at \$300 and the pitch still reads "alerts over GSM or LoRa".

- WWT-PRB-001, WWT-PRC-001 and WWT-REQ-001 are revised to v0.3. The key design choices in the precis are no longer "proposed". No requirement target changes as a result of these decisions; the 5 NTU and 0.2 mg/L defaults in R5 are now decided rather than proposed.
- The TRL 3 calculations (WWT-CAL-001) led to four design changes within these decisions: a smaller flow cell, a pressure-compensating flow regulator, a 30 s settling wait and a ventilated sun shield. They add \$12 to the parts cost, now \$282.
- WWT-CAL-001 shows that R1 is not met with D2's site pH at pH 7.5 and above. A relaxed R1 target and a trigger for the pH probe variant are proposed in `docs/REVIEW.md` and await Amish; this record does not decide them.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.